

Intermediate Elective 2

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Explaining my Inspiration

I chose Steven Ponces' "Coal falls. Renewables rise" as my visualization template because I liked how simplistic and sleek his representation of energy data is. This style makes sense for the data I plan to present, because it charts 2 variables (coal vs. renewable energy) that use the same units over the same time in history. I plan to chart % coverage of algae and % coverage of coyote-thistle in the same fashion, to see how the two variables follow or separate from each other. From the veg dataset, I'll be using num_quad and percent_cover to calculate a mean_cover for both algae and coyote-thistle, which will be on my y-axis, and I'll use the date column for the x-axis to show change over time.

1. Packages

```
## 1. LOAD PACKAGES & SETUP ----
suppressPackageStartupMessages({
  pacman::p_load(
    tidyverse, ggtext, showtext,
    janitor, scales, glue,
    camcorder, here, magick # <-- just add magick here
  )
})
```

2. Data

```
veg <- read_csv(here("data", "veg.csv"))
```

Rows: 27036 Columns: 17

-- Column specification -----

Delimiter: ","

chr (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...

dbl (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Rows: 257 Columns: 4

-- Column specification -----

Delimiter: ","

chr (2): year_pool, transect_name

dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

Necessary Code from Class

```
# creating new object from veg
pools_explicit0 <- veg %>%
  # filtering for vernal pools at ncos sites
  filter(str_detect(transect_name, "VP") &
         site == "ncos") %>%
  # handling missing data and filling with 0 values
  complete(year, transect_name, psoc, transect_distance,
           fill = list(percent_cover = 0)) %>%
  # grouping by year, transect_name, and species
  group_by(year, transect_name, psoc) %>%
  # calculating percent cover per year
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) %>%
  #ungrouping the df
```

```

ungroup() %>%
# joining with metadata to get year_pool
unite("year_pool", year, transect_name, na.rm = TRUE) %>%
left_join(metadata, by = "year_pool") %>%
# making a new column of mean percent cover
mutate(mean_pc = sum_pc/num_quad) %>%
# grouping by year and species
group_by(year, psoc) %>%
# calculating mean percent cover per year across pools
summarize(mean_pc_year = mean(mean_pc, na.rm = TRUE)) %>%
#ungrouping
ungroup()

```

`summarise()` has grouped output by 'year', 'transect_name'. You can override using the `.groups` argument.

`summarise()` has grouped output by 'year'. You can override using the `.groups` argument.

```

# filtering to investigate only coyote-thistle and algae
filtered_explicit0 <- pools_explicit0 %>%
  filter(psoc == "Eryngium_vaseyi" | psoc == "algae")

```

3. Tidy Data

```

# retrieving the peak value of algae cover
algae_peak <- filtered_explicit0 %>%
  filter(psoc == "algae") %>%
  slice_max(mean_pc_year)

# defining the year when algae peaked
algae_peak_year <- algae_peak$year
# defining the value of peak algae
algae_peak_val <- round(algae_peak$mean_pc_year, 1)

```

4. Visualization Parameters

```
### /- plot aesthetics ----
colors <- list(palette = list(
  bg_color      = "#0D1B2A",
  coyote_gold   = "#F4A932",
  algae_green    = "#009021",
  text_white    = "#FFFFFF",
  text_muted    = "#8A98A8",
  text_sub      = "#B0B8C4",
  grid_color    = "#1A2E42",
  accent_coral  = "#E8503A"
))

bg_color      <- colors$palette$bg_color
coyote_gold   <- colors$palette$coyote_gold
algae_green   <- colors$palette$algae_green
text_white    <- colors$palette$text_white
text_muted    <- colors$palette$text_muted
text_sub      <- colors$palette$text_sub
grid_color    <- colors$palette$grid_color
accent_coral  <- colors$palette$accent_coral
col_coyote    <- coyote_gold
col_algae     <- algae_green

### /- titles and caption ----
title_text <- glue(
  "<span style='font-weight:900; color:{algae_green};'>Algae
  ↪ Falls.&nbsp;</span>",
  "<span style='font-weight:900; color:{coyote_gold};'> Coyote-thistle
  ↪ persists.</span>"
)

subtitle_text <- glue(
  "<span style='font-size:12pt; font-weight:400; color:{text_sub};'>",
  "Despite mirroring the trajectory of coyote-thistle's percent cover<br>",
  "at NCOS vernal pools, algae sees a major dropoff in 2024 and 2025",
  "</span>"
)

caption_text <- "Source: Cheadle Center for Biodiversity and Ecological
  ↪ Restoration"
```

```

### font ----
# Load fonts directly - no dependency on his utils/fonts.R
font_add_google("Oswald", "title_font")
font_add_google("Roboto", "text_font")
showtext_auto()
showtext_opts(dpi = 320)

# Create a simple fonts list to match his structure
fonts <- list(
  title = "title_font",
  text  = "text_font"
)

```

5. Plot

```

### main plot ----
p <- filtered_explicit0 |>
  ggplot(aes(x = year, y = mean_pc_year, color = psoc, group = psoc)) +

  # Geom
  geom_hline(
    yintercept = seq(0, 10, by = 2),
    color = grid_color,
    linewidth = 0.25
  ) +
  geom_line(
    data = filtered_explicit0 |> filter(psoc == "Eryngium_vaseyi"),
    linewidth = 1.9,
    alpha = 1.0
  ) +
  geom_line(
    data = filtered_explicit0 |> filter(psoc == "algae"),
    linewidth = 1.3,
    alpha = 0.60
  ) +
  geom_point(
    data = filtered_explicit0 |> filter(year == 2021),
    size = 3.5,
    shape = 21,

```

```

    fill = bg_color,
    stroke = 1.8
) +
geom_point(
  data = filtered_explicit0 |> filter(year == 2025),
  size = 3.5,
  shape = 21,
  fill = bg_color,
  stroke = 1.8
) +

# Scales
scale_color_manual(
  values = c("Eryngium_vaseyi" = coyote_gold, "algae" = col_algae)
) +
scale_x_continuous(
  limits = c(2021, 2025),
  breaks = c(2021, 2022, 2023, 2024, 2025),
  expand = expansion(mult = c(0.03, 0.24))
) +
scale_y_continuous(
  limits = c(0, 10),
  breaks = seq(0, 10, by = 2),
  labels = \(x) glue("{x}%"),
  expand = expansion(mult = c(0.01, 0.04))
) +

# Annotate
annotate(
  "segment",
  x = algae_peak_year, xend = algae_peak_year,
  y = 0, yend = algae_peak_val,
  color = accent_coral,
  linewidth = 0.5,
  linetype = "dotted"
) +
annotate(
  "segment",
  x = 2025, xend = 2025,
  y = 4.45, yend = 9.45,
  color = "#D8E4EE",
  linewidth = 0.45,
  linetype = "dotted"
)

```

```

) +
annotate(
  "text",
  x = 2025,
  y = 6.5,
  label = "5% difference\nin pct cover ",
  hjust = 0,
  size = 4,
  color = "#D8E4EE",
  family = fonts$text,
  lineheight = 1.1
) +
annotate(
  "text",
  x = 2021.3,
  y = 1,
  label = glue("Algae"),
  hjust = 0,
  vjust = 0.5,
  size = 3.2,
  color = algae_green,
  family = fonts$text,
  lineheight = 1.15
) +
annotate(
  "text",
  x = 2021.3,
  y = 4,
  label = glue("Coyote-thistle"),
  hjust = 0,
  vjust = 0.5,
  size = 3.2,
  color = coyote_gold,
  family = fonts$text,
  lineheight = 1.15
) +
annotate(
  "text",
  x = 2022.7,
  y = 9,
  label = glue("Algae diverges from\n Coyote-thistle"),
  hjust = 0,
  vjust = 0.5,

```

```

    size = 3,
    color = accent_coral,
    family = fonts$text,
    lineheight = 1.15
) +

# Labels
labs(
  title = title_text,
  subtitle = subtitle_text,
  caption = caption_text,
  x = NULL,
  y = NULL
) +

# Theme
theme_void() +
theme(
  # canvas
  plot.background = element_rect(fill = bg_color, color = NA),
  panel.background = element_rect(fill = bg_color, color = NA),
  plot.title = element_markdown(
    family = fonts$title,
    size = 28,
    color = text_white,
    hjust = 0,
    lineheight = 1.1,
    margin = margin(t = 10, b = 6)
  ),
  plot.subtitle = element_markdown(
    family = fonts$text,
    size = 10,
    color = text_sub,
    hjust = 0,
    lineheight = 1.4,
    margin = margin(b = 22)
  ),
  plot.caption = element_markdown(
    family = fonts$text,
    size = 6.5,
    color = "#5A6878",

```

```

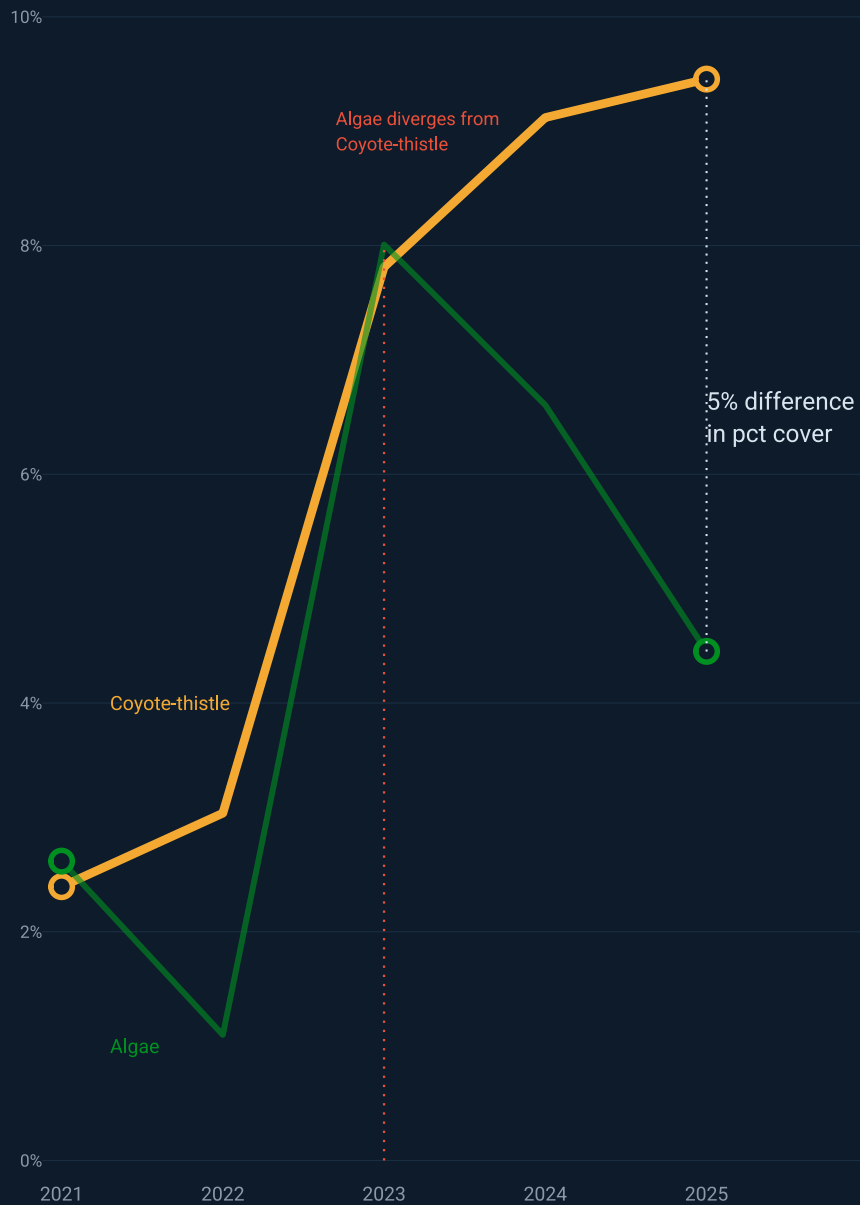
    hjust = 1,
    margin = margin(t = 14)
  ),
  axis.text.x = element_text(
    family = fonts$text,
    size = 9,
    color = text_muted,
    margin = margin(t = 6)
  ),
  axis.text.y = element_text(
    family = fonts$text,
    size = 8,
    color = text_muted,
    hjust = 1
  ),
  axis.line = element_blank(),
  axis.ticks = element_blank(),
  panel.grid = element_blank(),
  legend.position = "none",
  plot.margin = margin(t = 20, r = 16, b = 16, l = 20)
)

```

p

Algae Falls. Coyote-thistle persists.

Despite mirroring the trajectory of coyote-thistle's percent cover at NCOS vernal pools, algae sees a major dropoff in 2024 and 2025



Source: Cheadle Center for Biodiversity and Ecological Restoration